

16 Analyzing proportions and Goodness-of-fit

Example: A species of crocodile nests in sandy riverbanks, and the sex of the hatchlings is determined by the temperature of the surrounding sand during incubation. Historically, typical nest depths and shading produced an approximately 50:50 male-female sex ratio. Recently, hotter seasons and reduced vegetation cover are suspected to be warming the sand and shifting the sex ratio.

Researchers recorded the sex of 180 hatchlings emerging from sandy nests along a river in northern Australia. In their sample, 54 hatchlings were male and 126 were female.

What is the estimate of the proportion of female crocodiles in this population?

The sample proportion:

$$\hat{p} = \frac{x}{n} = \frac{126}{180} = 0.7$$

$n = 180$ $p = ?$
 $\hat{p} = .7$

How precise is this estimate?

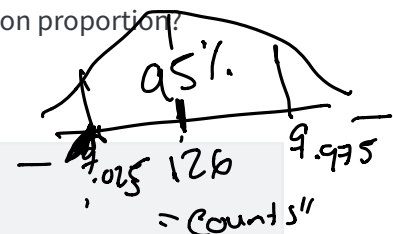
We can measure precision using the standard error of the estimated proportion:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 0.034$$

What is the “exact” 95% confidence interval of the population proportion?

In our example:

```
lower<-qbinom(.025,180,.7)/180
upper<-qbinom(.975,180,.7)/180
print(paste("95% CI is: (",lower,upper,""))
```



$Bin(n, p)$
 $n = 180$
 $p = .7$

mean $np = 126$ [1] "95% CI is: (0.633333333333333 0.766666666666667)" exact C.I

An alternative approximate CI is

$$\hat{p} \pm 1.96 SE(\hat{p})$$

normal approximation
C.I.

That is:

```
phat<-126/180
se<-sqrt(phat*(1-phat)/180)
lower<-phat-1.96*se
upper<-phat+1.96*se
```

```
print(paste("95% CI is: (", lower, upper, ")"))
```

```
[1] "95% CI is: ( 0.633053254995731 0.766946745004269 )"
```

where 1.96 is the 0.975 quantile of the standard normal distribution.

However a common approximation is the Agresti-Coull approximation that estimates the population proportion by

$$\tilde{p} = \frac{x + \frac{1}{2}}{n + 4} = \frac{128}{184} = 0.696$$

better approx when n is small

with

$$SE_{\tilde{p}} = \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}} = 0.033$$

*"shrinkage"
"Bayesian like" correction.*

and 95% confidence interval:

$$\tilde{p} \pm 1.96 SE_{\tilde{p}}$$

```
n<-180
tilde_p<-128/184
se<-sqrt(tilde_p*(1-tilde_p)/(n+4))
print(paste("95% CI is: (", tilde_p-1.96*se, tilde_p+1.96*se, ")"))
```

```
[1] "95% CI is: ( 0.629166460204054 0.762137887622033 )"
```

16.1 Binomial test

Example 2: A study of 25 genes involved in spermatogenesis found their locations in the mouse genome. The study was carried out to test a prediction of evolutionary theory that such genes should occur disproportionately often on the X chromosome. As it turned out, 10 out of 25 spermatogenesis genes (40%) were on the X chromosome. If genes for spermatogenesis occurred "randomly" throughout the genome, then we would expect only 6.1% of them to fall on the X chromosome, because the X chromosome contains 6.1% of the genes in the genome. Do the results support the hypothesis that spermatogenesis genes occur preferentially on the x chromosome?

$H_0 : p = 0.061$, the probability that a spermatogenesis gene falls on the X chromosome is .061.

$H_A : p \neq 0.061$ (Two-sided)

$n = 25$ $P_{null} = .061$

```
pval<-2*(1-pbinom(9,25,.061))
pval ~ 2 P(X ≥ 10) = 2 (1 - P(X ≤ 9))
```

$pbinom(x, \dots) = P(X \leq x)$



```
[1] 1.987976e-06
```

This p-value is below the significance level of $\alpha = .05$, therefore, we reject the null hypothesis and conclude that there is a disproportionate number of spermatogenesis genes on the X chromosome.

16.2 Goodness of fit test

The binomial test just introduced is an example of goodness-of-fit test. A **goodness-of-fit test** is a method for comparing an observed frequency distribution with the frequency distribution that would be expected under a simple probability model governing the occurrence of different outcomes. In particular, we will use a test statistic called χ^2 (chi squared) calculated as follows:

$$\chi^2 = \sum_i \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

Where Observed_i is the frequency of individuals observed in the i -th category and Expected_i is the frequency expected in that category under the null hypothesis.

Example: the proportional model

We would expect that babies should be born at the same frequency on all seven days of the week. To test this, we obtained the number of babies born on each day of the week in a random sample of 180 births from 2016.

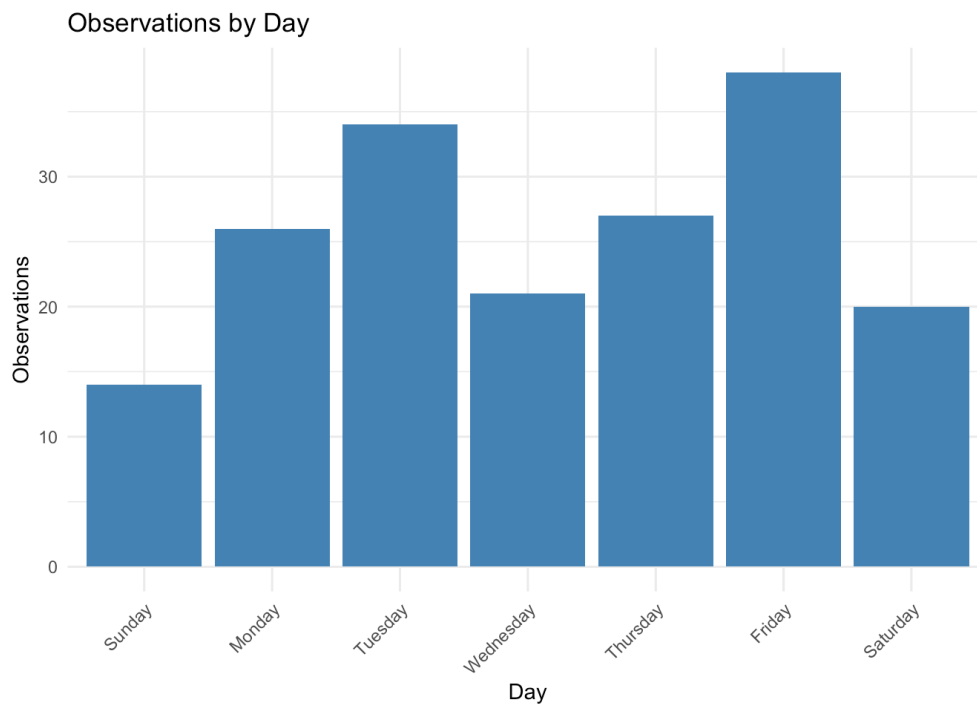
The observe data

```
day<-c("Sunday","Monday","Tuesday","Wednesday","Thursday","Friday",
obs<-c(14,26,34,21,27,38,20) → 180 babies
obs<-obs/sum(obs)
df <- data.frame(day = factor(day, levels = day), obs = obs)
library(ggplot2)

ggplot(df, aes(x = day, y = obs)) +
  geom_col(fill = "steelblue") +
  labs(x = "Day", y = "Observations", title = "Observations by Day") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

exp for Sunday. $\frac{52}{366} \times 180$

exp = $\left(\frac{52}{366} \times 180, \frac{52}{366} \times 180, \dots, \frac{53}{366} \times 180 \right)$



Under the proportional model (null hypothesis), the number of births on Monday is proportional to the number of Mondays in 2016. H_0 : The frequency of births on each day of the week is proportional to the number of times that day occurs in the year

H_A : The frequency of births on each day of the week is NOT proportional to the number of times that day occurs in the year

The expected proportion under the null is then:

```
exp<-c(52,52,52,52,52,53,53)*180/366

chi_square<-sum ((obs-exp)^2 / exp)
chi_square
```

[1] 15.79465 ↩

The null distribution of the χ^2 statistic is well approximated by the theoretical χ^2 distribution with degrees of freedom calculated as follows:

✕ $df = \text{Num. of categories} - 1 = \text{Num. of parameters estimated from the data}$

```
# Chi-square distributions with df = 10 and df = 20

df_vals <- c(2, 10)

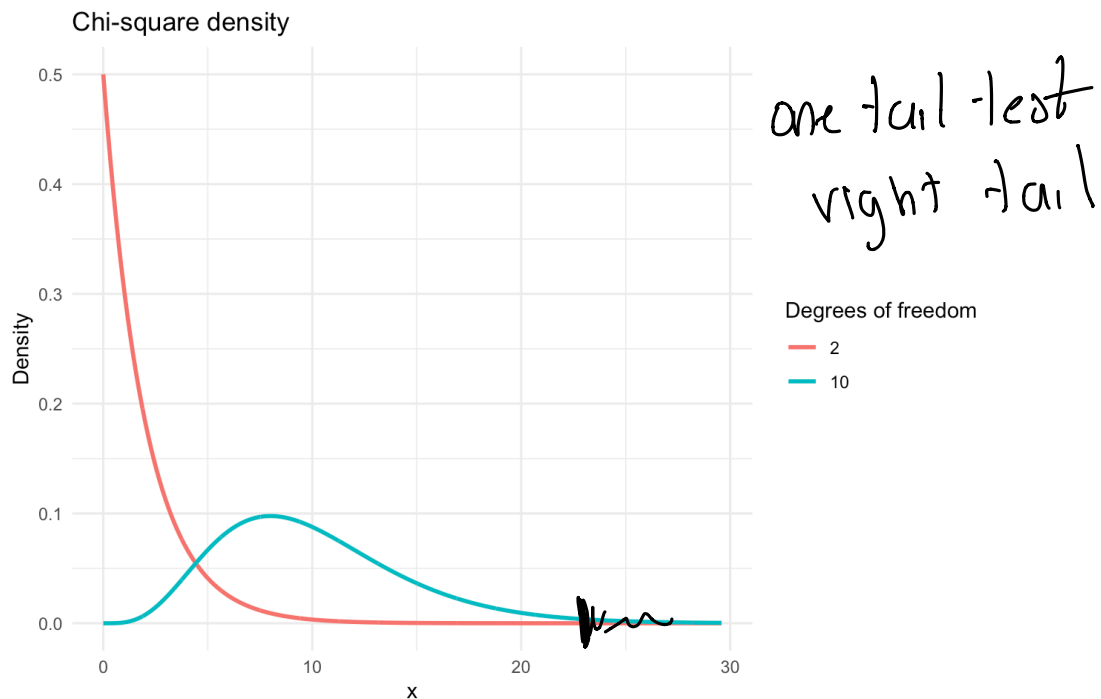
# x-range covering essentially all mass for both
x <- seq(0, qchisq(0.999, df = max(df_vals)), length.out = 2000)

library(ggplot2)

dat <- data.frame(
```

```
x = rep(x, times = length(df_vals)),
df = factor(rep(df_vals, each = length(x))),
density = c(dchisq(x, df = 2), dchisq(x, df = 10))
)

ggplot(dat, aes(x = x, y = density, color = df)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Chi-square density",
    x = "x",
    y = "Density",
    color = "Degrees of freedom"
  ) +
  theme_minimal()
```



Calculating the p-value

For the χ^2 goodness-of-fit test, the p-value is the probability of getting a χ^2 value **greater** than the observed test value calculated from the data (why?).

```
pval<-1-pchisq(chi_square,df=6)
pval
```

$P(X \geq \text{chi-square})$
 $= 1 - P(X \leq \text{chi-square})$

```
[1] 0.01489962
```

At the standard significance level of $\alpha = 0.05$, such a small p-value leads us to reject the null hypothesis.

Assumptions:

- Individuals in the data set are a random sample from the whole population

- None of the categories should have an expected frequency less than one
- No more than 20% of the categories should have expected frequencies less than five

If one of these conditions is not met, then we can combine some of the categories with small expected frequencies.

16.3 Goodness-of-fit when there are 2 categories

Example: migraine headache

Patients who suffered from moderate to severe migraine headache took part in a double-blind clinical trial to assess an experimental surgery. A group of 75 patients were randomly assigned to receive either the real surgery on migraine trigger sites (n=49) or a sham surgery (n=26) in which an incision was made but no further procedure was performed. The surgeons hoped that patients would experience a substantial reduction in migraine headaches.

```
# Contingency table: rows = (Success, No success), cols = (Real, Sham)
tab <- matrix(
  c(41, 8,
    15, 11),
  nrow = 2, byrow = FALSE,
  dimnames = list(
    Outcome = c("Success", "No success"),
    Group = c("Real", "Sham")
  )
)

#tab
addmargins(tab) # show row/column totals
```

Expected

$$S \frac{49 \cdot 56}{75} \quad 26 \cdot \frac{56}{75}$$

$$N \frac{49 \cdot (1 - \frac{56}{75})}{75} \quad 26 \cdot (1 - \frac{56}{75})$$

$$\frac{49}{75} \quad \frac{26}{75}$$

	Group		
Outcome	Real	Sham	Sum
Success	41	15	56
No success	8	11	19
Sum	49	26	75

Expected.

$\frac{56}{75}$ overall success.

one tail / per P-value calculation

The focus of interest in a **contingency table** is the dependence or association between the column variable and the row variable (between treatment and response). Under the null hypothesis, there is no such association.

$$H_0 : P(\text{Success} \mid \text{Real}) = P(\text{Success} \mid \text{Sham})$$

$$H_A : P(\text{Success} \mid \text{Real}) \neq P(\text{Success} \mid \text{Sham}) \quad \leftarrow \text{two-sided.}$$

Under the null, the estimated success rate is $56/75 = 0.746$ and so, the expected values are:

$$49 \times \frac{56}{75}$$

```
exp<-matrix(0,nrow=2,ncol=2)
exp[1,]<-c(49*56/75,26*56/75)
exp[2,]<-c(49,26)-exp[1,]
exp
```

$\exp[1,1] \leftarrow$
 $\exp[1,2] \leftarrow$

```
      [,1]      [,2]
[1,] 36.58667 19.41333
[2,] 12.41333  6.58667
```

Then, the chi-square statistic is

```
chi<-sum(((exp-tab)^2)/exp)
chi
```

```
[1] 6.061864
```

```
pval<-1-pchisq(chi,df=1)
pval
```

```
[1] 0.01381318
```

Thus, we reject the null and find that the data provide sufficient evidence to conclude that the real surgery is better than the sham surgery for reducing migraine headache.

16.4 Independence and association

Example: hair color and eye color

To study the relationship between hair color and eye color in a German population, an anthropologist observed a sample of 6800 men, with the following results:

```
tab <- matrix(
  c(726, 3129,
    131, 2814),
  nrow = 2, byrow = FALSE,
  dimnames = list(
    Eye_Color = c("Dark", "Light"),
    Hair_Color = c("Dark", "Light")
  )
)

#tab
addmargins(tab)
```

	Hair_Color		
Eye_Color	Dark	Light	Sum
Dark	726	131	857
Light	3129	2814	5943

Expected under null

$$\frac{857}{6800} \times 3855$$

$$\frac{857}{6800} \times 2945$$

Sum 3855 2945 6800

$$\frac{5943}{6800} \times 3855 \quad \frac{5943}{6800} \times 2945$$

H_0 : Eye color is independent of hair color

H_A : Eye color and is not independent of hair color

```
exp<-matrix(0,nrow=2,ncol=2)
exp[1,]<-c(3855*857/6800,2945*857/6800)
exp[2,]<-c(3855,2945)-exp[1,]
exp
```

```
      [,1]      [,2]
[1,] 485.8434 371.1566
[2,] 3369.1566 2573.8434
```

```
chi<-sum(((exp-tab)^2)/exp)
chi
```

```
[1] 313.6315
```

```
pval<-1-pchisq(chi,df=1)
pval
```

```
[1] 0
```

We reject the null hypothesis.

16.5 The Chi-Square test for the $r \times k$ table

The goal of statistical analysis of an $r \times k$ contingency table is to investigate the relationship between the row and column variables. In this situation, the expected frequencies are calculated as

$$e = \frac{(\text{Row total}) \times (\text{Column total})}{\text{Grand total}}$$

Critical values for the chi-square test are obtained with

$$df = (r - 1) \times (k - 1)$$

Example: Plover nesting

Wildlife ecologists monitored the breeding habitats of mountain plovers for three years and made note of where the plovers nested. They found 66 nests on agricultural fields (AF), 67 nests in shortgrass prairie dog habitat (PD), and 20 nests on other grassland (G). The nesting choices varied across the years for these 153 sampled plover broods. The observed data is:

```
tab <- matrix(
```



```

c(21, 19, 26,
  17, 38, 12,
  5, 6, 9),
nrow = 3, byrow = TRUE,
dimnames = list(
  Location = c("Agricultural field (AF)", "Prairie dog habitat",
  Year = c("2004", "2005", "2006")
)
)

#tab
addmargins(tab)

```

Location	Year			Sum
	2004	2005	2006	
Agricultural field (AF)	21	19	26	66
Prairie dog habitat (PD)	17	38	12	67
Grassland (G)	5	6	9	20
Sum	43	63	47	153

$e = 63 \times \frac{66}{153}$

The expected frequencies are then

```

exp<-matrix(0,nrow=3,ncol=3)
sumrow<-apply(tab,1,sum)
sumcol<-apply(tab,2,sum)
sumrow

```

```

Agricultural field (AF) Prairie dog habitat (PD)
Grassland (G)
20 66 67

```

```
sumcol
```

```

2004 2005 2006
43 63 47

```

```

exp[,1]<-sumrow/sum(sumrow)*sumcol[1]
exp[,2]<-sumrow/sum(sumrow)*sumcol[2]
exp[,3]<-sumrow/sum(sumrow)*sumcol[3]

```

```
exp
```

```

      [,1]      [,2]      [,3]
[1,] 18.549020 27.176471 20.274510
[2,] 18.830065 27.588235 20.581699
[3,]  5.620915  8.235294  6.143791

```

$$df = (r-1)(c-1) = 2 \times 2 = 4$$

```

chi<-sum(((exp-tab)^2)/exp)
chi

```

```
[1] 14.08935
```

```
pval<-1-pchisq(chi,df=4)  
pval
```

```
[1] 0.007015278
```

The chi-square shows that there is significant evidence that the nesting location preferences differed across the three years.

Shortcut:

```
chisq.test(tab) ``
```